1.0 Introduction

In an automobile, engine is the main source for developing the power and the power produced from it must be transmitted to the wheels of the vehicle to move it from one place to another. The system which is developed for transmitting the power from the engine to the wheels of the vehicle is called as the automobile transmission system or vehicle power transmission system. For basic needs of the vehicle and for space requirements, the manufacturers place the engines in appropriate locations and transmit the energy from the engine to the wheels by various ways. Let us discuss some of the ways how the power is transmitted to the wheels with suitable diagrams.

1.1 Transmission system

In automobiles the rotary power (or torque) is obtained from the engine. The developed power is transmitted to the rotating wheels on the road. The unit which has various components connected in between the engine and the rear wheels of the vehicle is called as the automobile power transmission system. Let us see the important parts of an automobile transmission system.

1.1.1 Parts of Transmission System

1. Clutch
2. Gear box
3. Propeller shaft
4. Universal joint
5. Differential
6. Rear axle



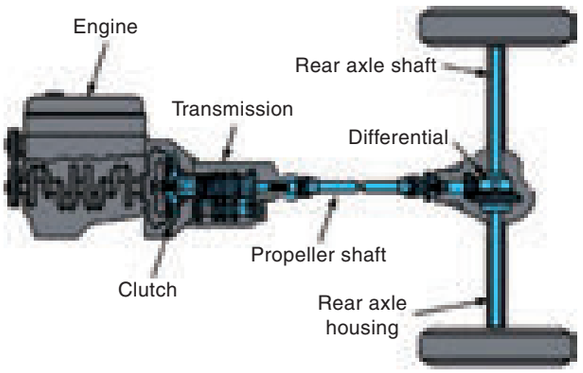


Figure 1.1.1 Important components of a transmission system of an automobile

1.2 Needs of transmission system

- For transmitting and stopping torque developed from the engine to the rear axle of the vehicle.
- For transmitting power to wheels without any noise during the engine is running.
- For avoiding vibrational effects during the transmission of power from engine to the rear wheels.
- For operating the vehicle wheels at the required speed.
- For balancing the speed of the wheels during vehicle is in running.

1.3 Types of transmission system

1. Front Engine Rear Wheel Drive
2. Front Engine Front Wheel Drive
3. Front Engine Four Wheel Drive
4. Rear Engine Rear Wheel Drive

1.3.1 Front engine rear wheel drive

Generally front engine rear wheel drive is commonly used in most of the vehicles. The power from the engine to

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